

**Nirma University
Institute of Technology**



Course Code :- 6CS103CC26

Course Name :- OS

Practical - 2

Student Name

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Student Roll No.

26MCA061

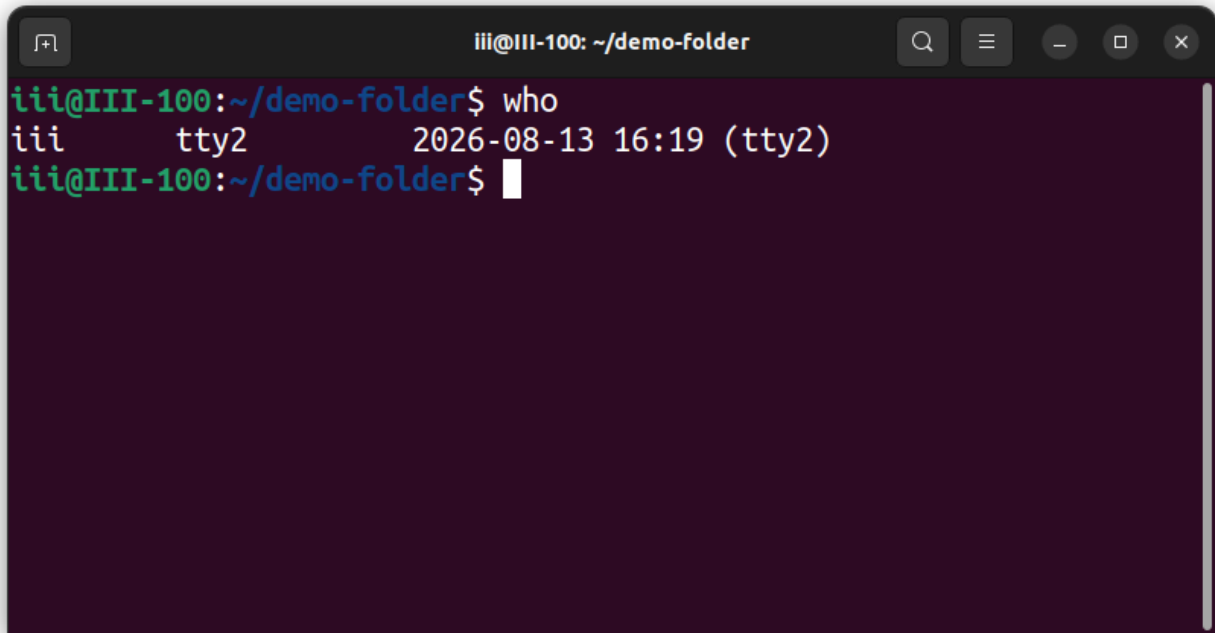
Faculty In Charge

Prof. Drashti Patel

UNIX Commands

1. who

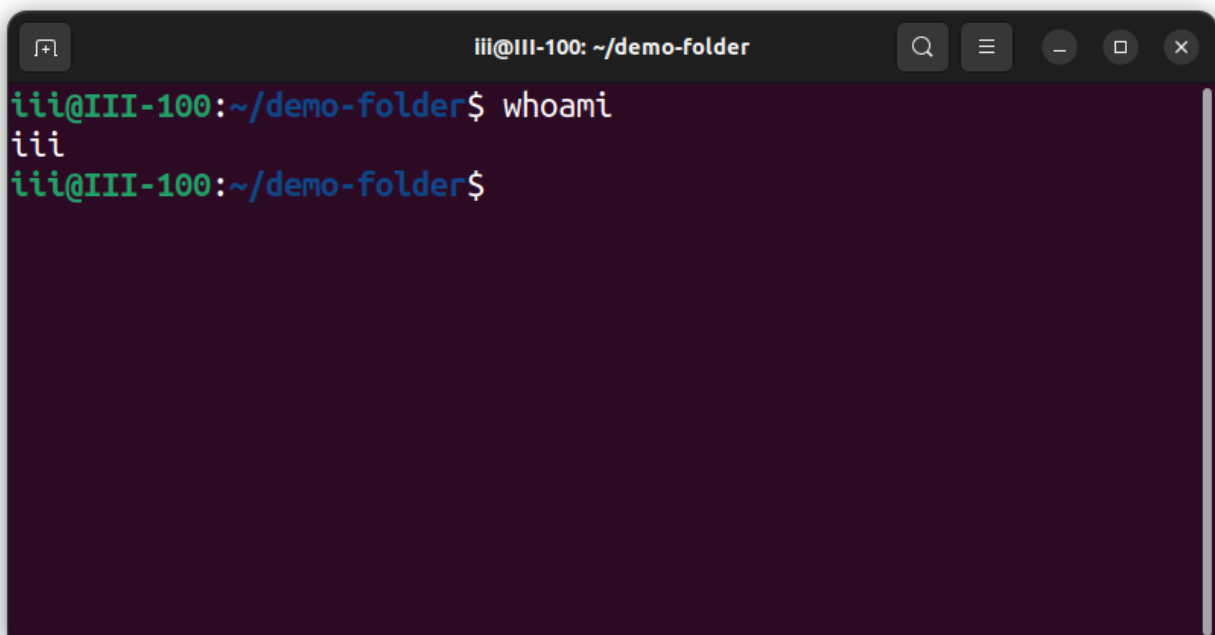
- Description: Shows users who are currently logged into the system.
- Example: who



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ who
iii      tty2      2026-08-13 16:19 (tty2)
iii@III-100:~/demo-folder$
```

2. whoami

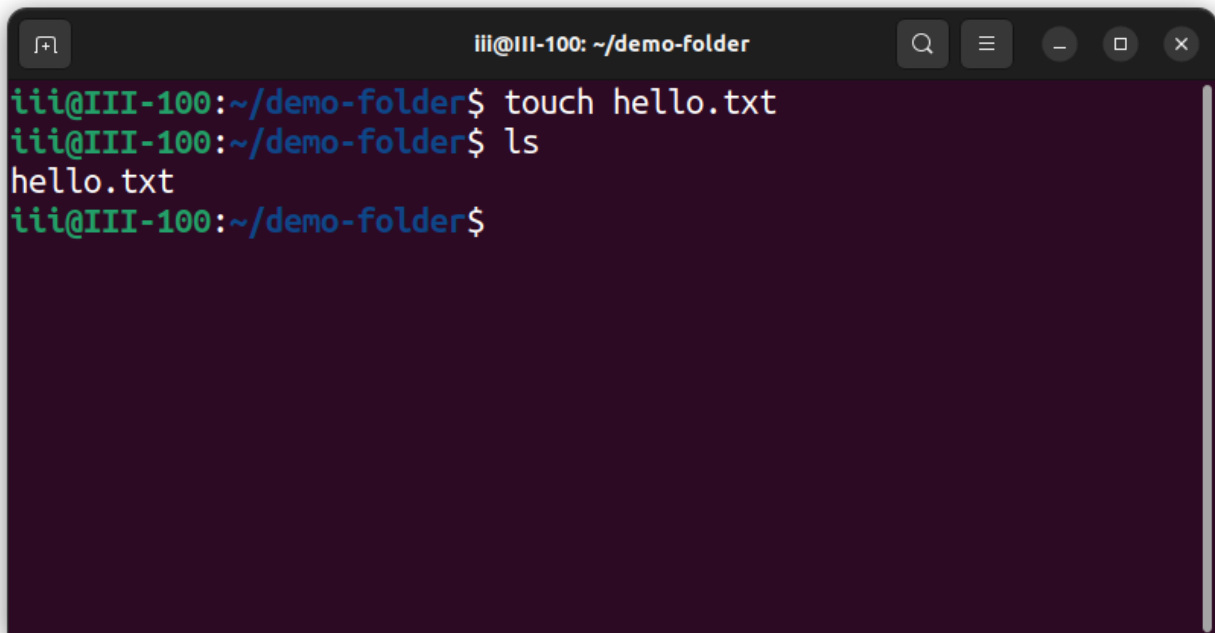
- Description: Displays the username of the current user.
- Example: whoami



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ whoami
iii
iii@III-100:~/demo-folder$
```

4. touch

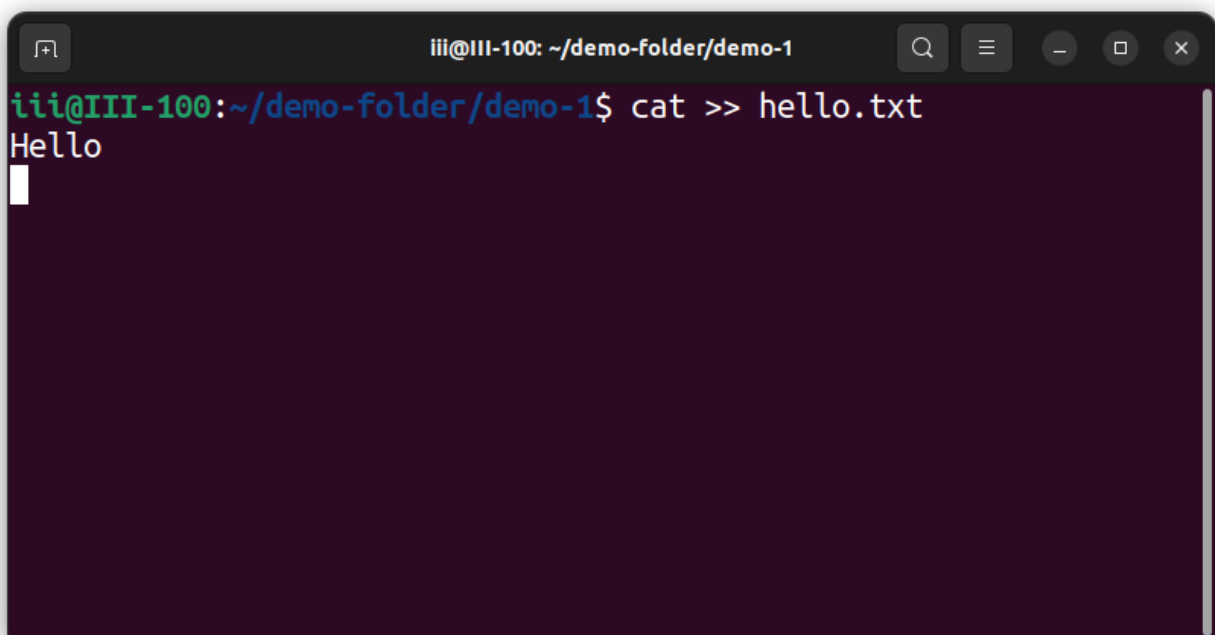
- Description: Creates an empty file or updates the timestamp of an existing file.
- Example: touch file.txt



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ touch hello.txt
iii@III-100:~/demo-folder$ ls
hello.txt
iii@III-100:~/demo-folder$
```

5. cat

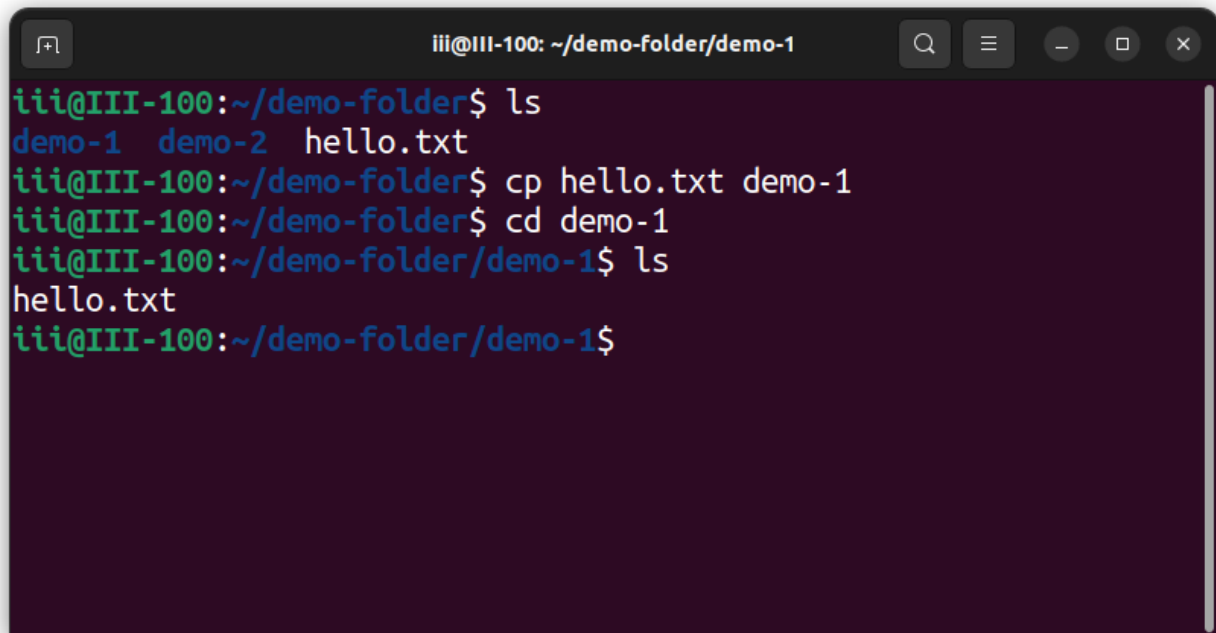
- Description: Displays or concatenates file contents. With redirection, it can create, overwrite, or append to a file.
- Example: cat file.txt
- Example: cat > file.txt



```
iii@III-100: ~/demo-folder/demo-1
iii@III-100:~/demo-folder/demo-1$ cat >> hello.txt
Hello
```

6. cp

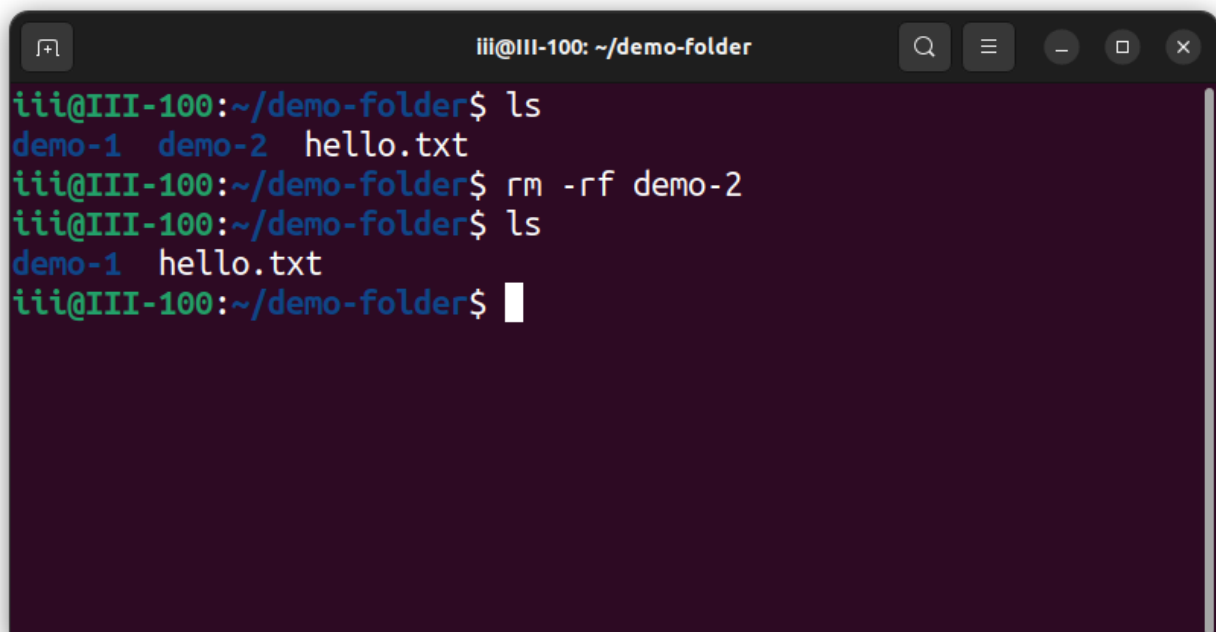
- Description: Copies files. Use -r to copy directories and their contents.
- Example: cp file1.txt file2.txt
- Example: cp -r dir1 dir2



```
iii@III-100: ~/demo-folder/demo-1
iii@III-100:~/demo-folder$ ls
demo-1  demo-2  hello.txt
iii@III-100:~/demo-folder$ cp hello.txt demo-1
iii@III-100:~/demo-folder$ cd demo-1
iii@III-100:~/demo-folder/demo-1$ ls
hello.txt
iii@III-100:~/demo-folder/demo-1$
```

7. rm

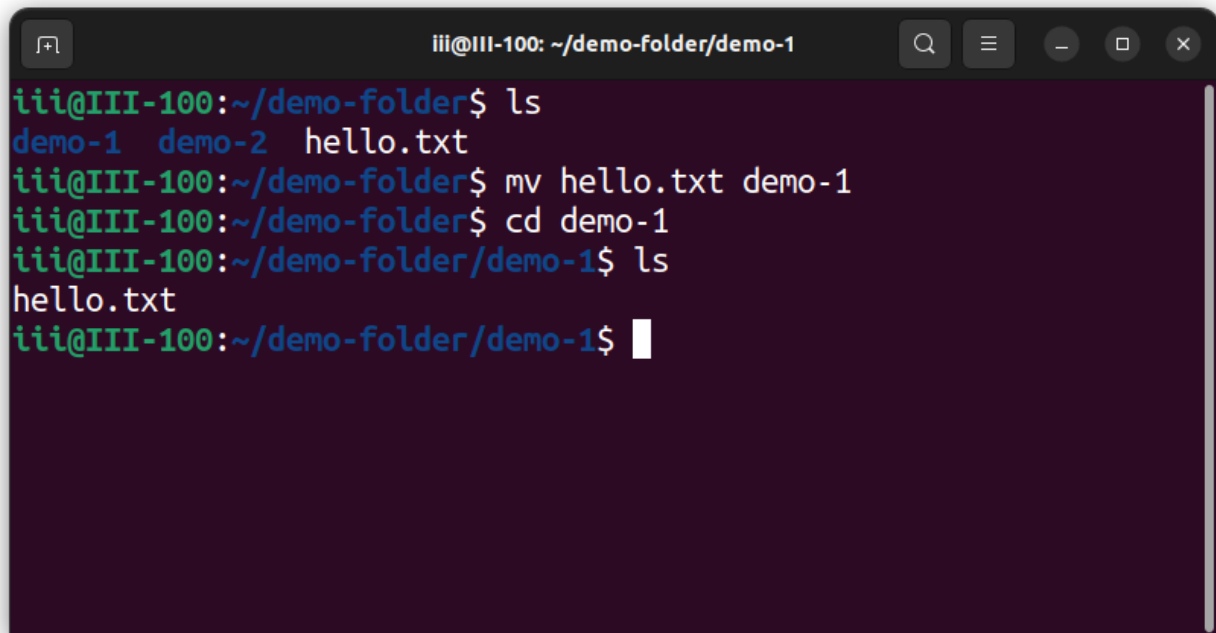
- Description: Removes files. Use -r to remove a directory and its contents.
- Example: rm file.txt
- Example: rm -r folder



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ ls
demo-1  demo-2  hello.txt
iii@III-100:~/demo-folder$ rm -rf demo-2
iii@III-100:~/demo-folder$ ls
demo-1  hello.txt
iii@III-100:~/demo-folder$
```

8. mv

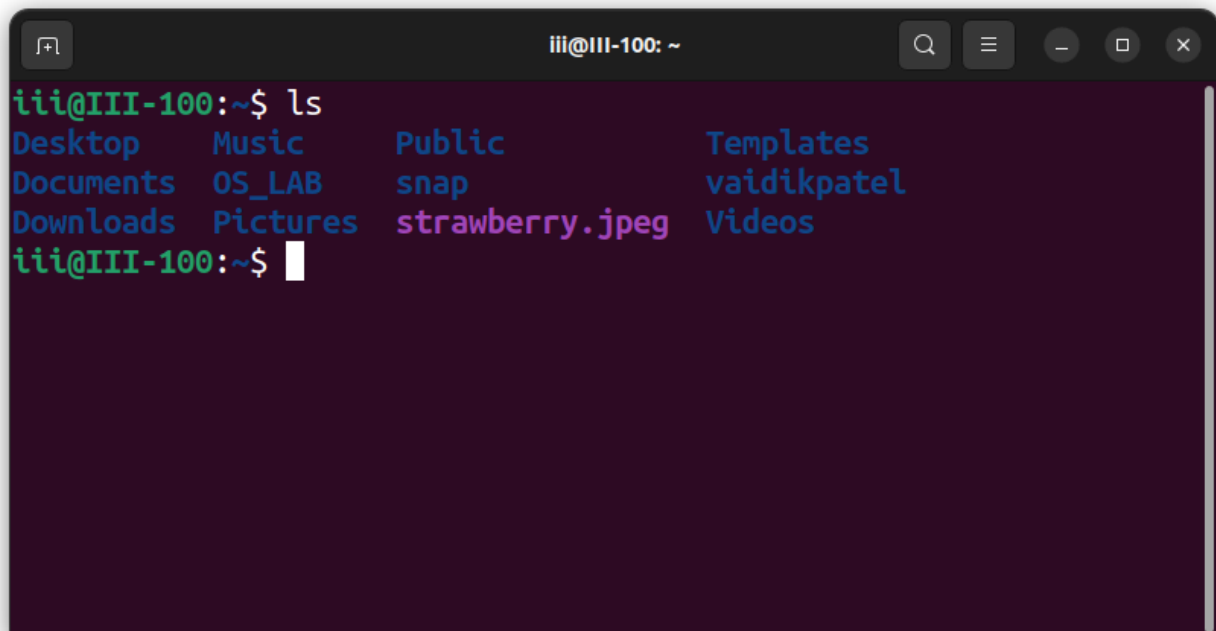
- Description: Moves or renames files or directories.
- Example: mv old.txt new.txt



```
iii@III-100: ~/demo-folder/demo-1
iii@III-100:~/demo-folder$ ls
demo-1  demo-2  hello.txt
iii@III-100:~/demo-folder$ mv hello.txt demo-1
iii@III-100:~/demo-folder$ cd demo-1
iii@III-100:~/demo-folder/demo-1$ ls
hello.txt
iii@III-100:~/demo-folder/demo-1$
```

9. ls

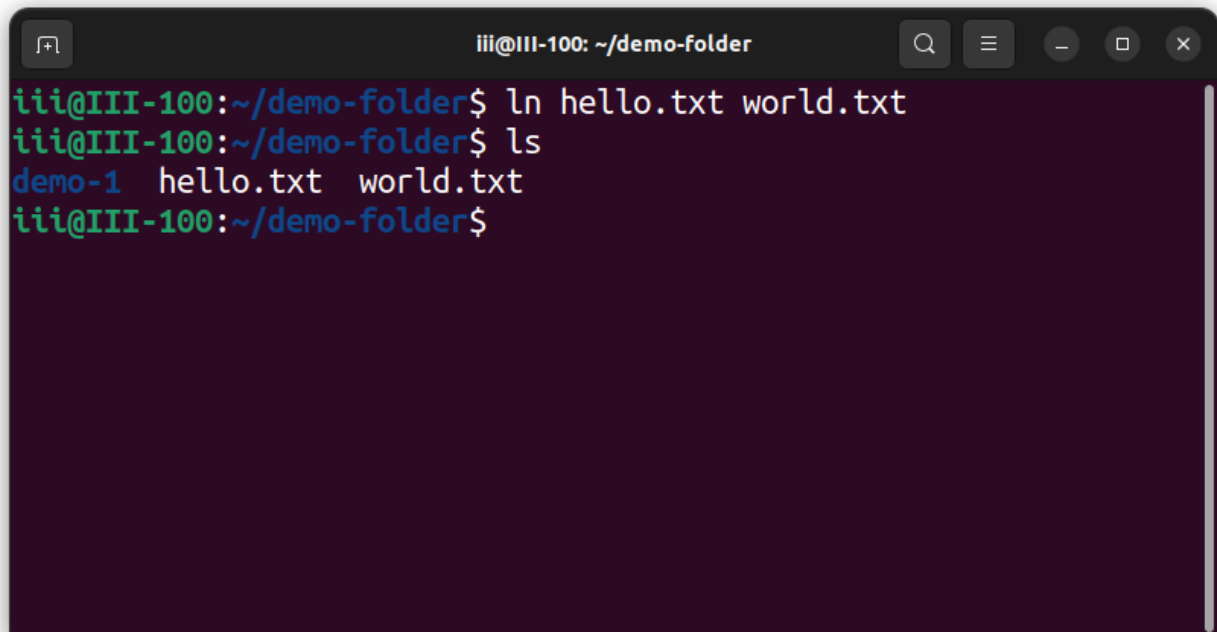
- Description: Lists files and directories.
- Example: ls -l



```
iii@III-100: ~
iii@III-100:~$ ls
Desktop    Music      Public      Templates
Documents  OS_LAB     snap        vaidikpatel
Downloads  Pictures   strawberry.jpeg  Videos
iii@III-100:~$
```

10. ln

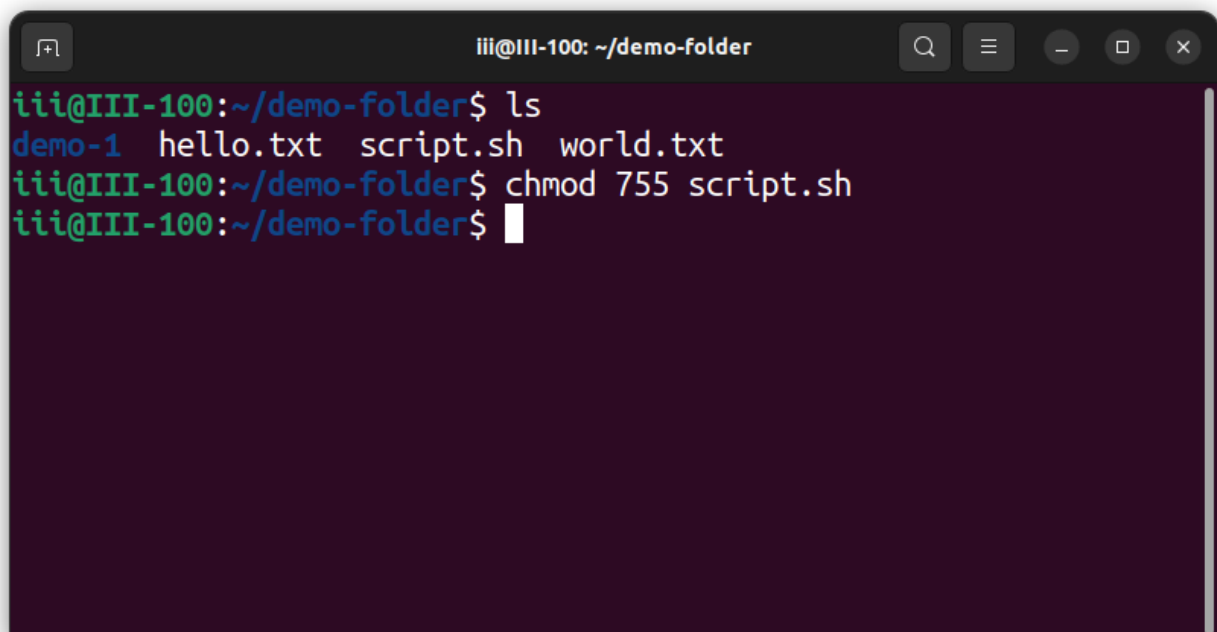
- Description: Creates links. By default it creates a hard link; -s creates a symbolic link.
- Example: ln file1 link1
- Example: ln -s target.txt link.txt

A terminal window titled 'iii@III-100: ~/demo-folder' with search, menu, and window control icons. The terminal shows the command 'ln hello.txt world.txt' being executed, followed by 'ls' which lists 'demo-1 hello.txt world.txt'.

```
iii@III-100:~/demo-folder$ ln hello.txt world.txt
iii@III-100:~/demo-folder$ ls
demo-1  hello.txt  world.txt
iii@III-100:~/demo-folder$
```

11. chmod

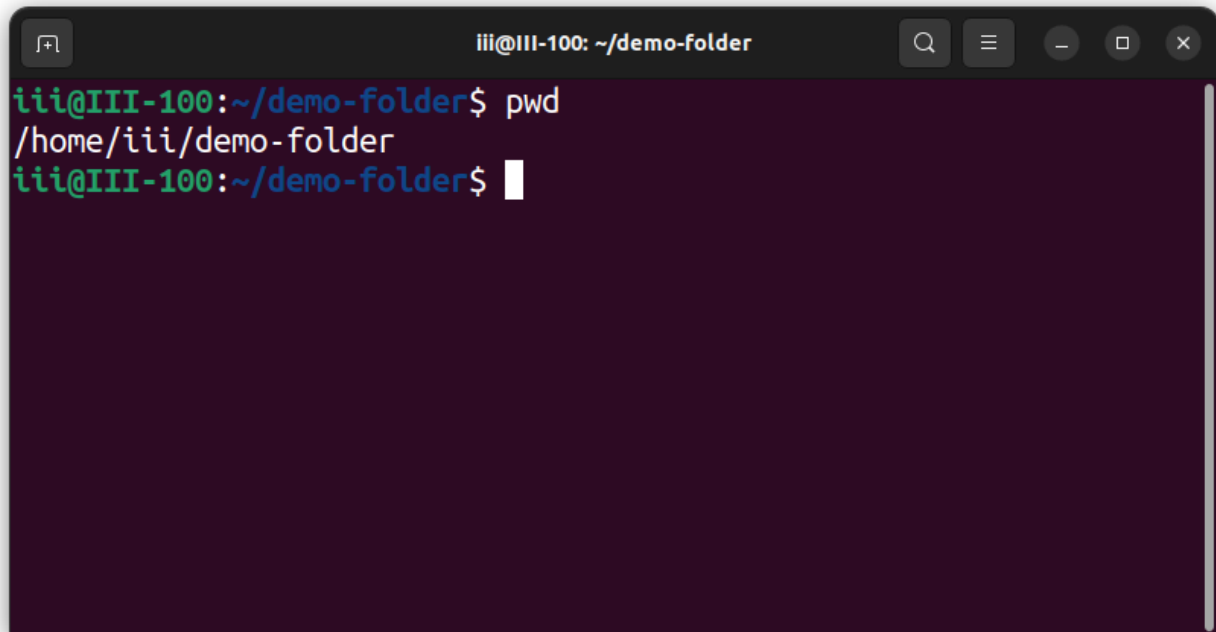
- Description: Changes file or directory permissions.
- Example: chmod 755 script.sh

A terminal window titled 'iii@III-100: ~/demo-folder' with search, menu, and window control icons. The terminal shows 'ls' listing 'demo-1 hello.txt script.sh world.txt', followed by 'chmod 755 script.sh' being executed, and a final prompt line.

```
iii@III-100:~/demo-folder$ ls
demo-1  hello.txt  script.sh  world.txt
iii@III-100:~/demo-folder$ chmod 755 script.sh
iii@III-100:~/demo-folder$
```

13. pwd

- Description: Prints the current working directory.
- Example: pwd

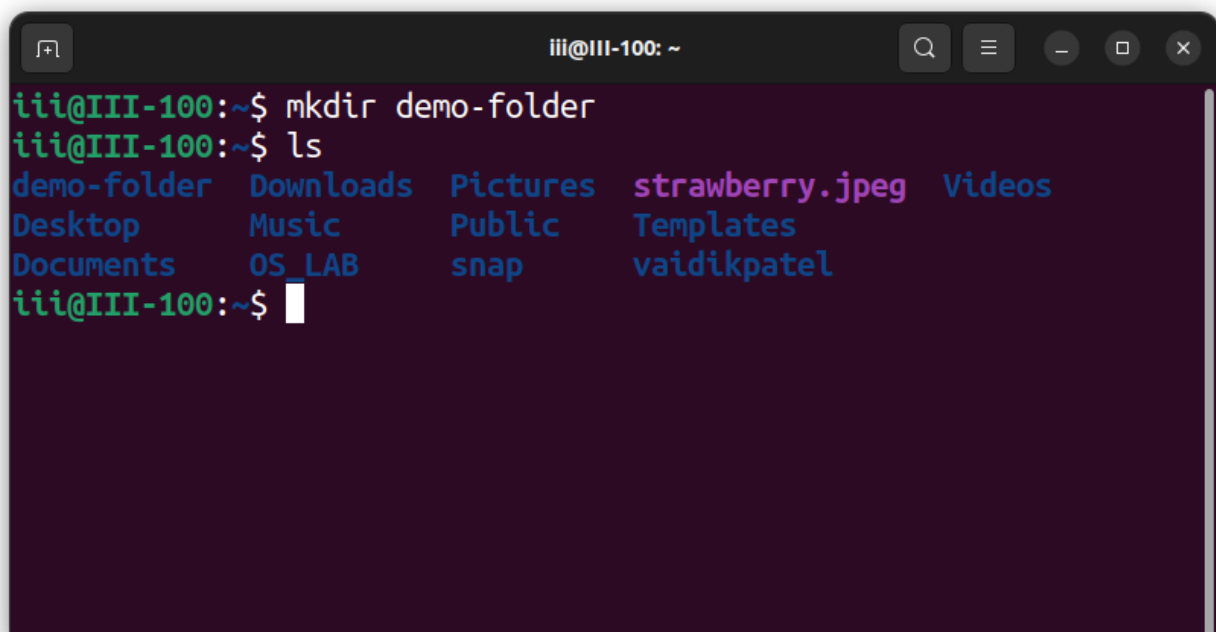


```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ pwd
/home/iii/demo-folder
iii@III-100:~/demo-folder$
```

A terminal window titled "iii@III-100: ~/demo-folder" with search, menu, and window control icons. The command `pwd` is entered, and the output is `/home/iii/demo-folder`.

14. mkdir

- Description: Creates a new directory.
- Example: `mkdir new_folder`

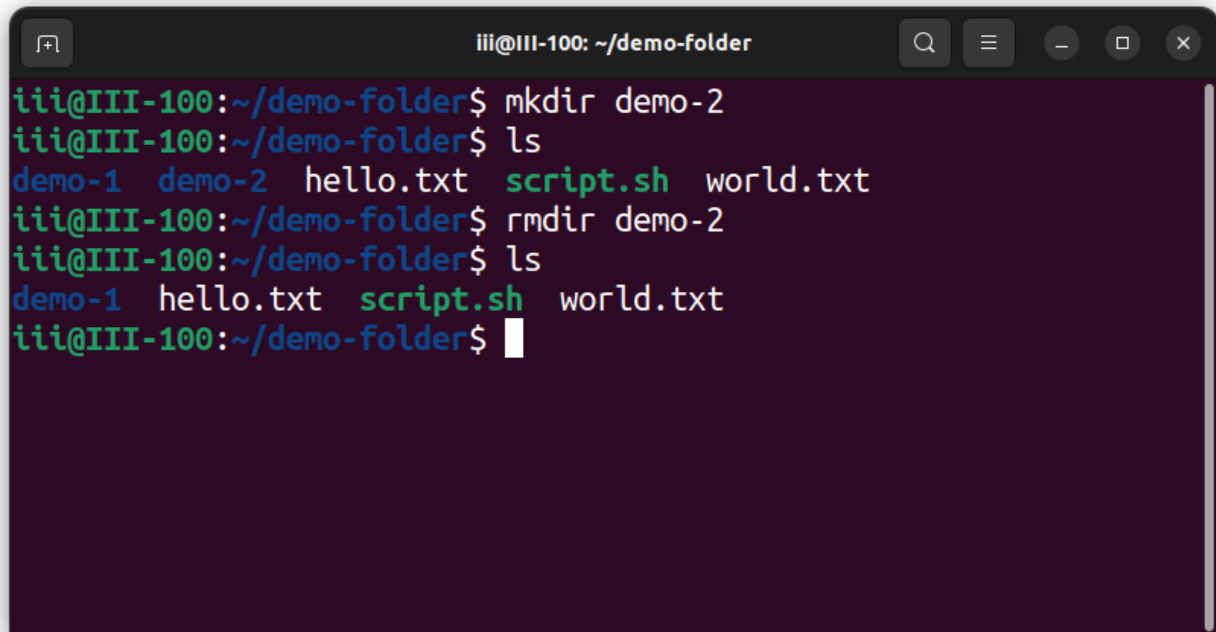


```
iii@III-100: ~
iii@III-100:~$ mkdir demo-folder
iii@III-100:~$ ls
demo-folder  Downloads  Pictures  strawberry.jpeg  Videos
Desktop      Music      Public    Templates
Documents    OS_LAB     snap      vaidikpatel
iii@III-100:~$
```

A terminal window titled "iii@III-100: ~" with search, menu, and window control icons. The command `mkdir demo-folder` is entered. Then, the `ls` command is entered, showing a list of files and directories including the newly created `demo-folder`.

15. rmdir

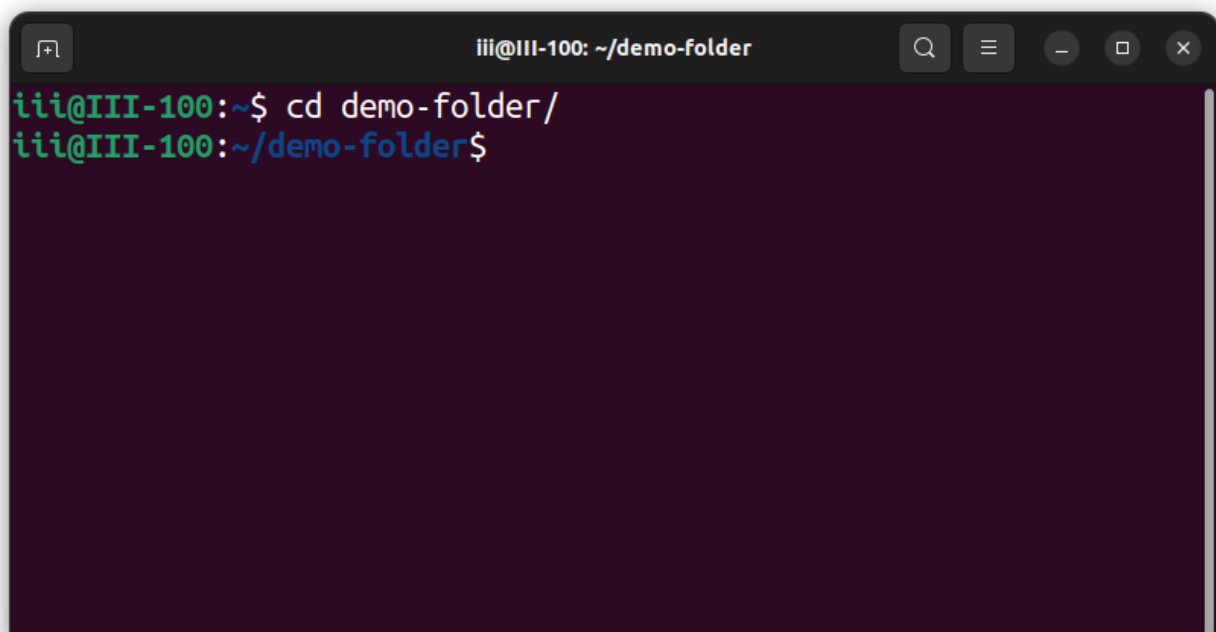
- Description: Removes an empty directory.
- Example: rmdir old_folder



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ mkdir demo-2
iii@III-100:~/demo-folder$ ls
demo-1  demo-2  hello.txt  script.sh  world.txt
iii@III-100:~/demo-folder$ rmdir demo-2
iii@III-100:~/demo-folder$ ls
demo-1  hello.txt  script.sh  world.txt
iii@III-100:~/demo-folder$
```

16. cd

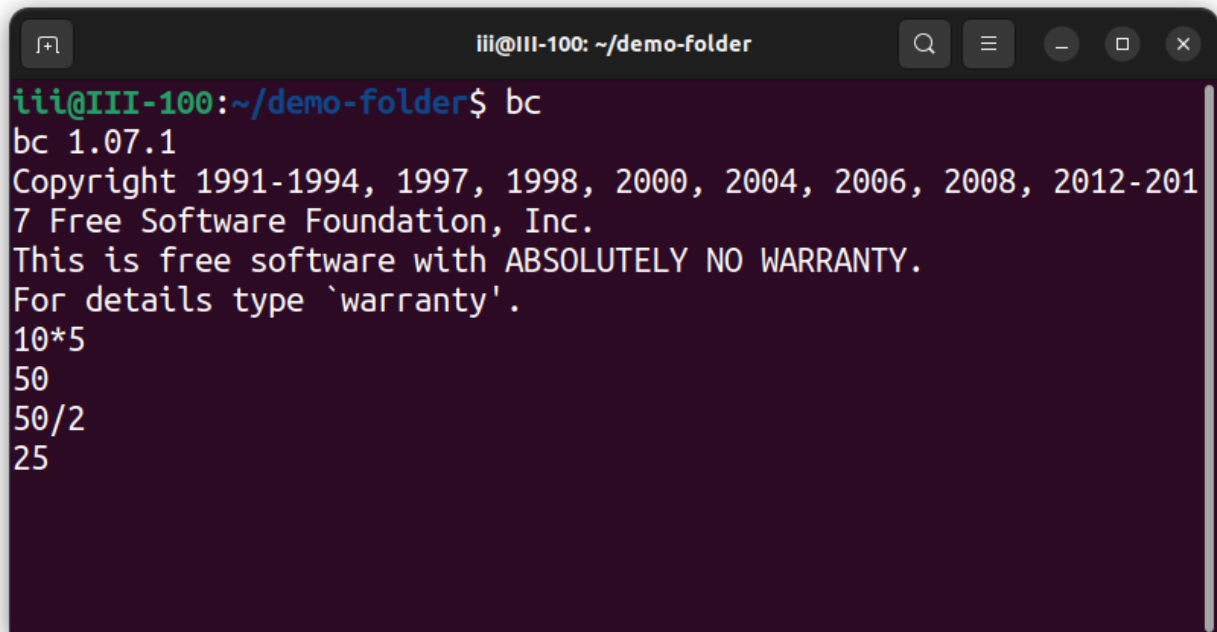
- Description: Changes the current working directory.
- Example: cd Documents



```
iii@III-100: ~/demo-folder
iii@III-100:~$ cd demo-folder/
iii@III-100:~/demo-folder$
```

17. bc

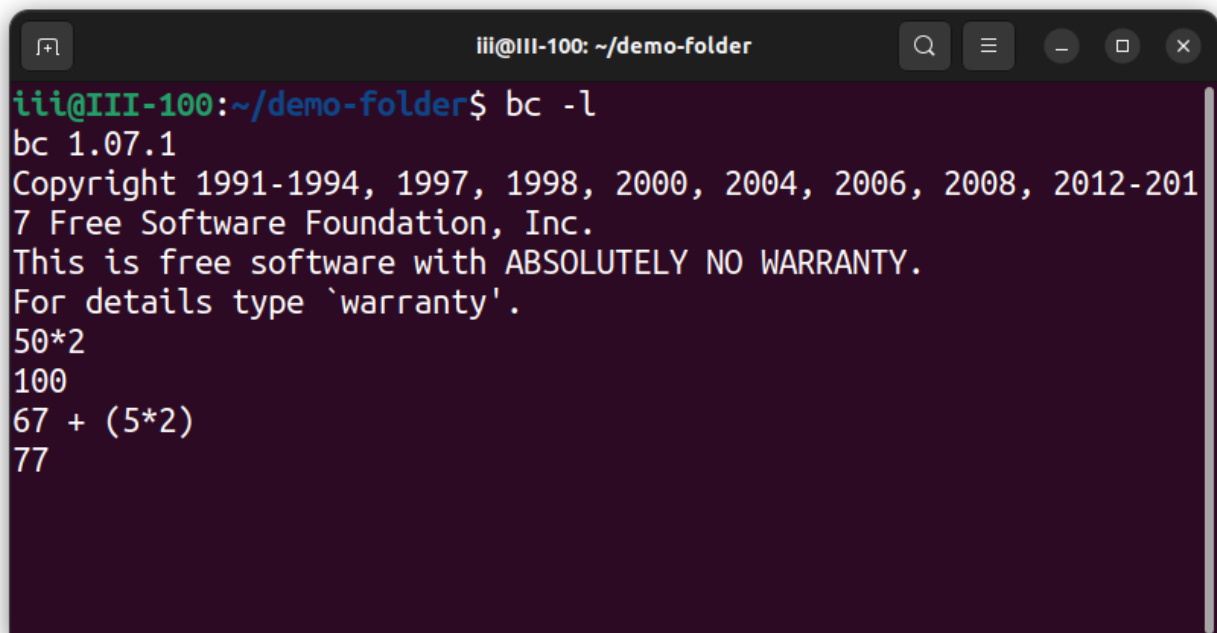
- Description: Launches an arbitrary-precision calculator.
- Example: bc



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-201
7 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
10*5
50
50/2
25
```

18. bc -l

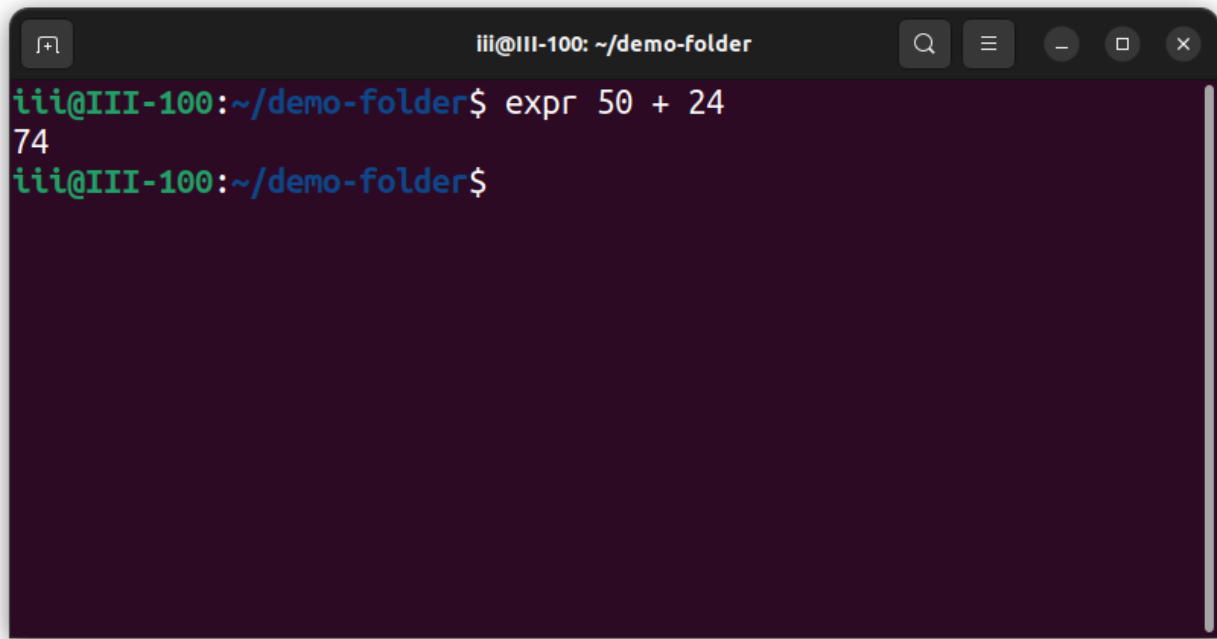
- Description: Launches bc with its math library enabled.
- Example: bc -l



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ bc -l
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-201
7 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
50*2
100
67 + (5*2)
77
```

19. expr

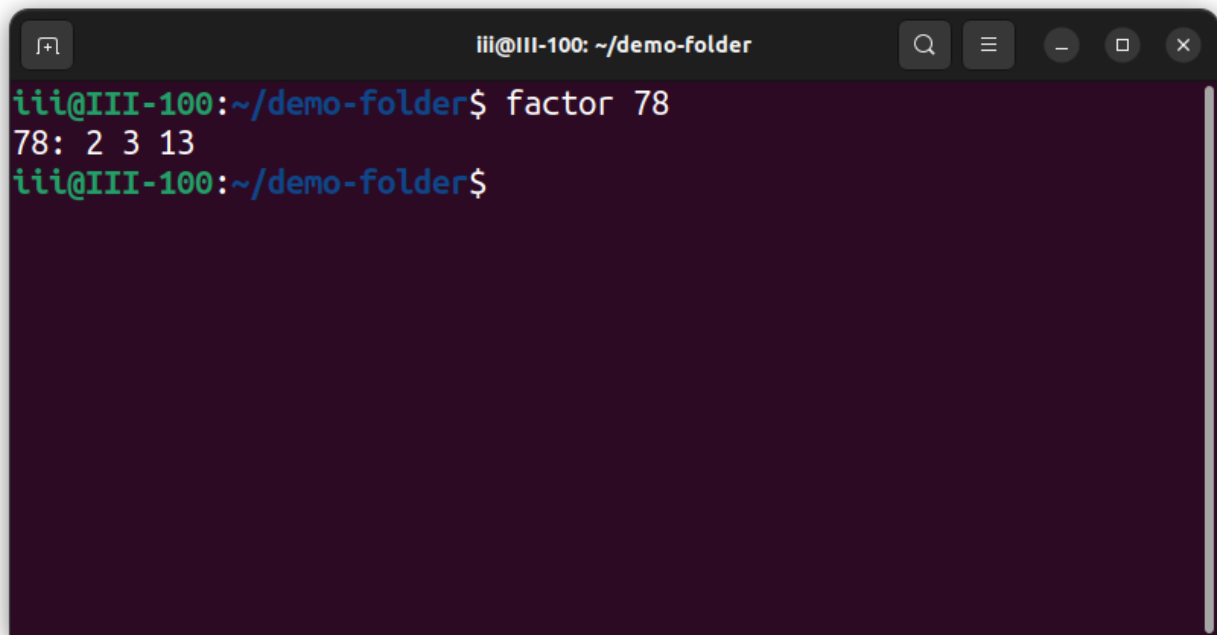
- Description: Evaluates expressions.
- Example: `expr 5 + 3`

A terminal window with a dark background and light-colored text. The window title is 'iii@III-100: ~/demo-folder'. The prompt is 'iii@III-100:~/demo-folder\$'. The user enters the command 'expr 50 + 24'. The output '74' is displayed on the next line. The prompt 'iii@III-100:~/demo-folder\$' is shown again on the following line.

```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ expr 50 + 24
74
iii@III-100:~/demo-folder$
```

20. factor

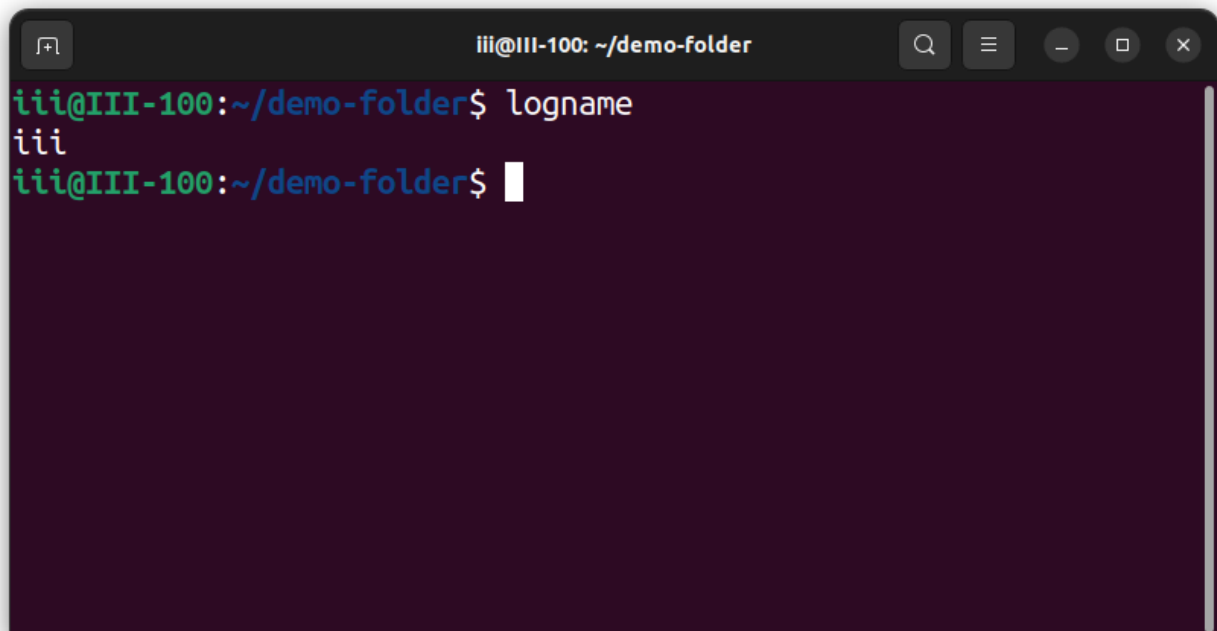
- Description: Finds the prime factors of a number.
- Example: `factor 60`

A terminal window with a dark purple background. The title bar shows 'iii@III-100: ~/demo-folder'. The prompt is 'iii@III-100:~/demo-folder\$'. The command 'factor 78' has been entered and executed, resulting in the output '78: 2 3 13'. The prompt is now 'iii@III-100:~/demo-folder\$' again.

```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ factor 78
78: 2 3 13
iii@III-100:~/demo-folder$
```

21. logname

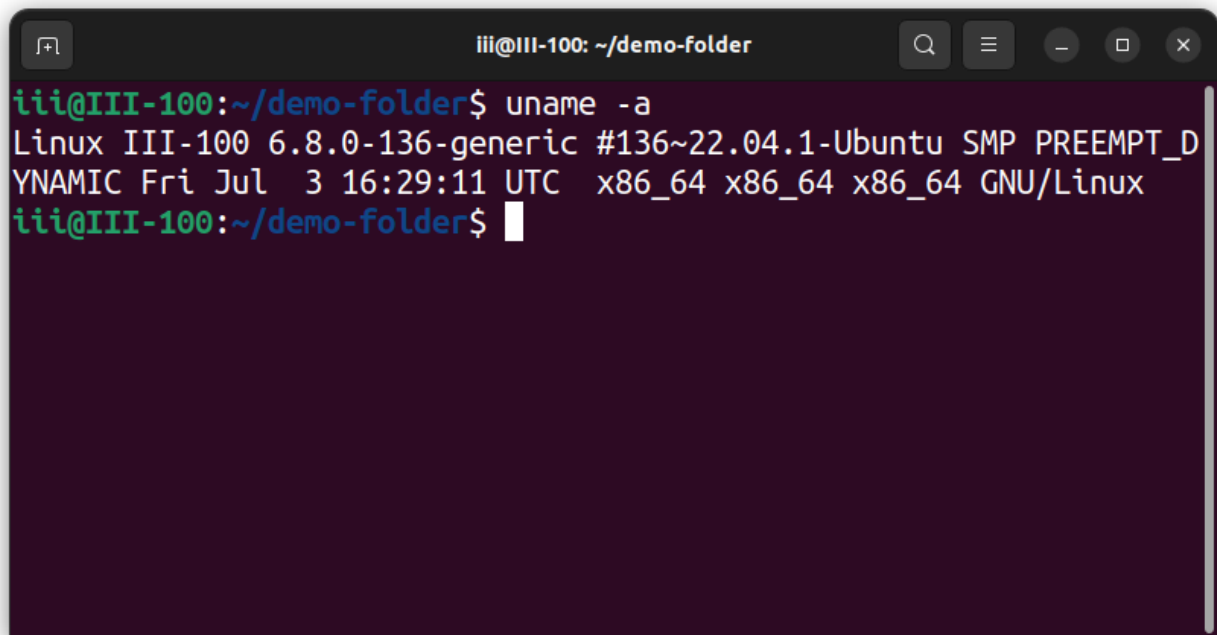
- Description: Displays the login name of the current user.
- Example: logname

A terminal window with a dark purple background. The title bar shows 'iii@III-100: ~/demo-folder'. The prompt is 'iii@III-100:~/demo-folder\$'. The command 'logname' has been entered and executed, resulting in the output 'iii'. The prompt is now 'iii@III-100:~/demo-folder\$' again.

```
iii@III-100:~/demo-folder$ logname
iii
iii@III-100:~/demo-folder$
```

22. uname

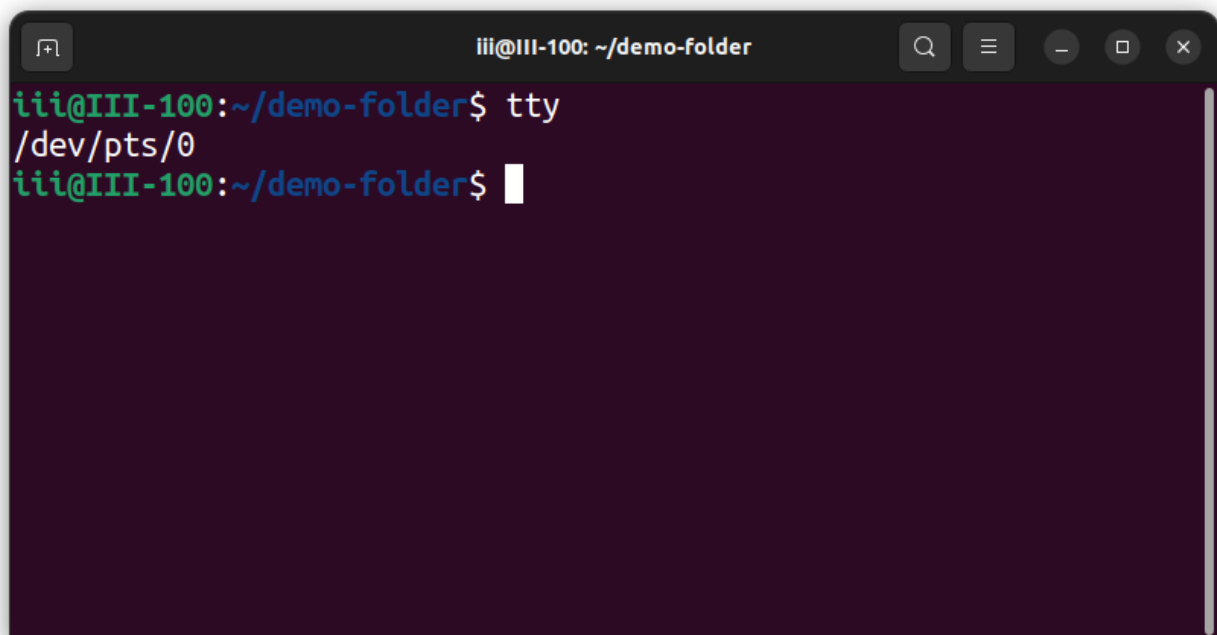
- Description: Prints system information.
- Example: uname -a



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ uname -a
Linux III-100 6.8.0-136-generic #136~22.04.1-Ubuntu SMP PREEMPT_D
YNAMIC Fri Jul 3 16:29:11 UTC x86_64 x86_64 x86_64 GNU/Linux
iii@III-100:~/demo-folder$
```

23. tty

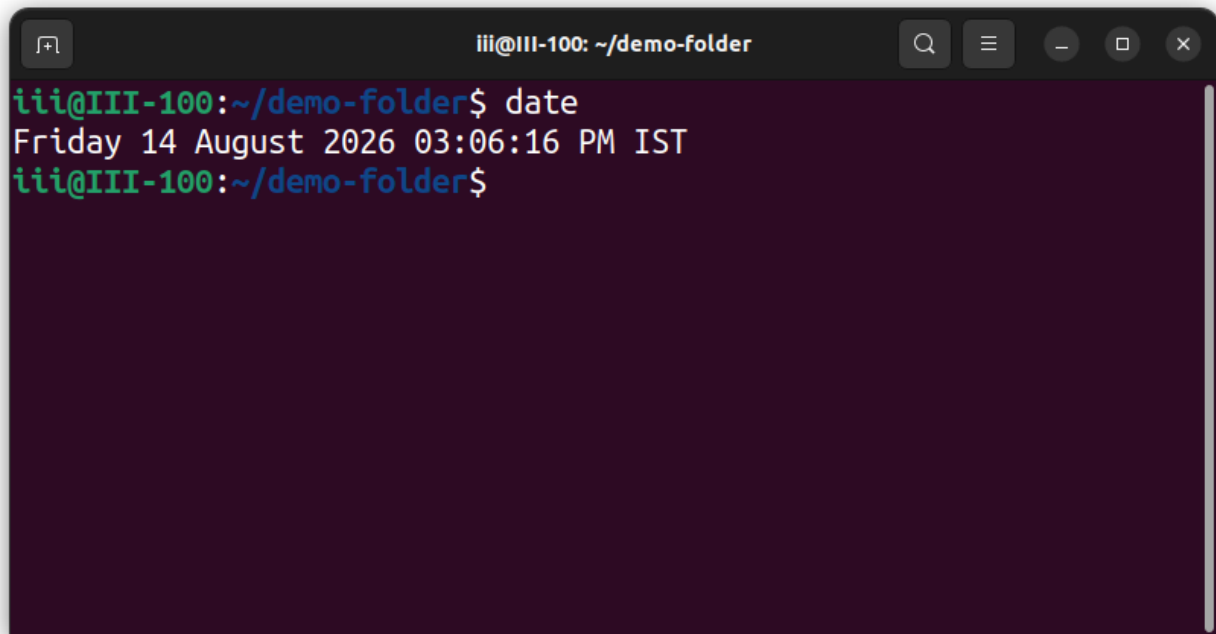
- Description: Shows the terminal device associated with the current session.
- Example: tty



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ tty
/dev/pts/0
iii@III-100:~/demo-folder$
```

24. date

- Description: Displays or sets the system date and time.
- Example: date
- Example: date "+%d-%m-%Y"

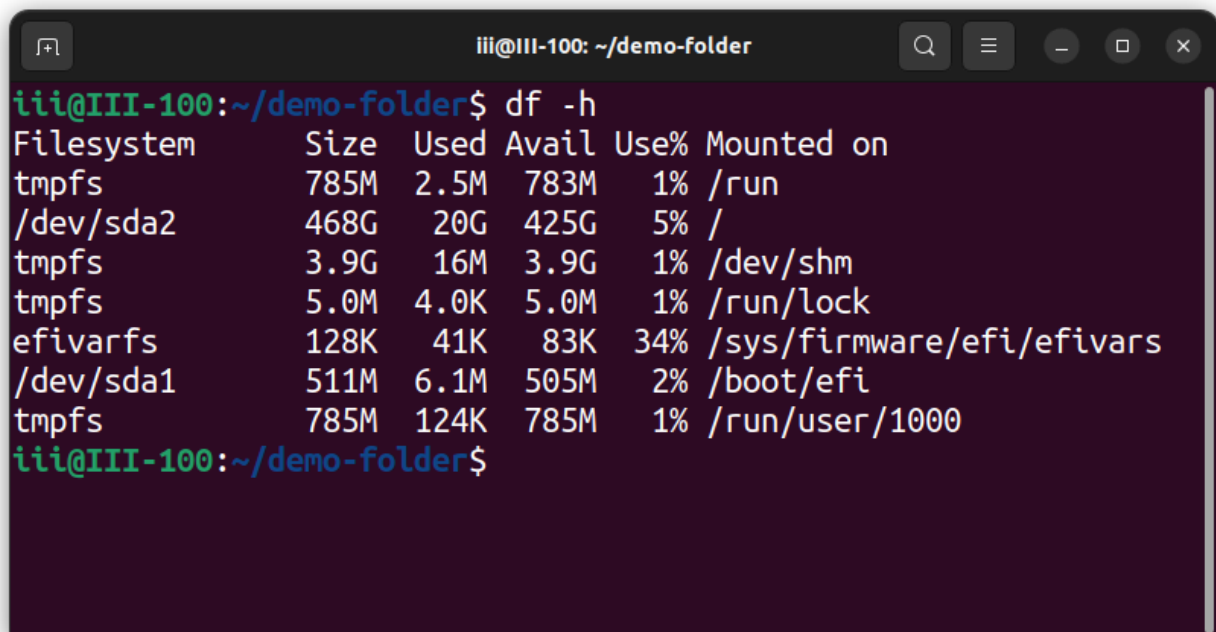


A terminal window titled 'iii@III-100: ~/demo-folder' showing the execution of the 'date' command. The output is 'Friday 14 August 2026 03:06:16 PM IST'.

```
iii@III-100: ~/demo-folder$ date
Friday 14 August 2026 03:06:16 PM IST
iii@III-100: ~/demo-folder$
```

25. df

- Description: Shows disk-space usage. -h displays values in human-readable units.
- Example: df -h

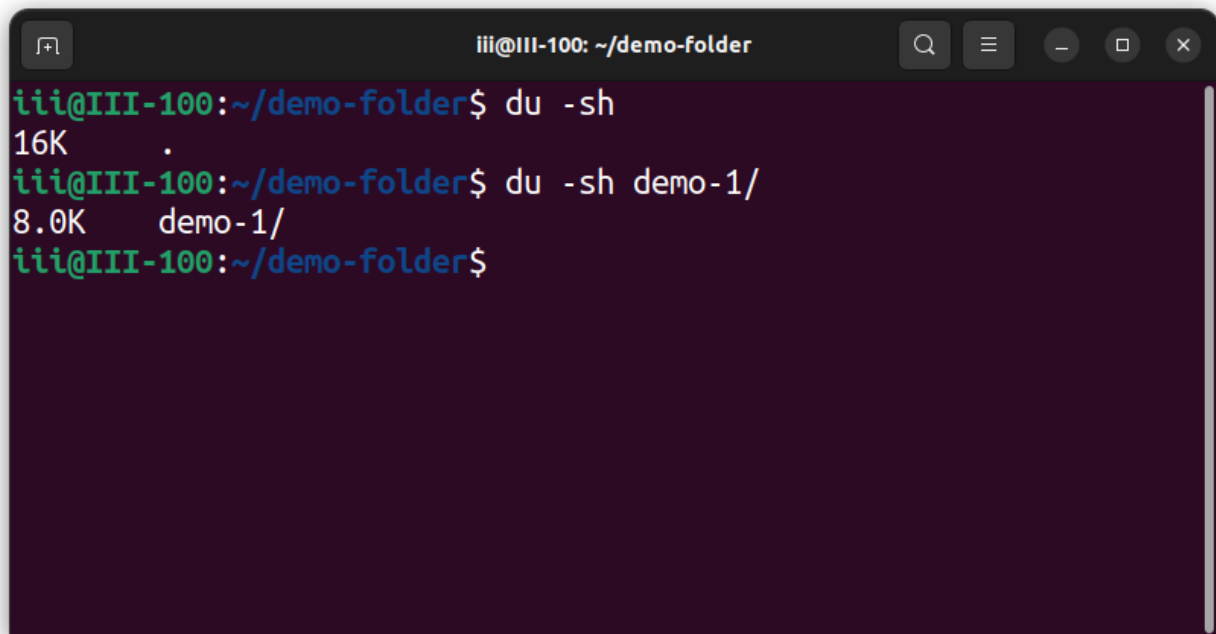


A terminal window titled 'iii@III-100: ~/demo-folder' showing the execution of the 'df -h' command. The output is a table of disk space usage for various filesystems.

```
iii@III-100: ~/demo-folder$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           785M  2.5M  783M   1% /run
/dev/sda2       468G   20G  425G   5% /
tmpfs           3.9G   16M   3.9G   1% /dev/shm
tmpfs           5.0M   4.0K   5.0M   1% /run/lock
efivarfs        128K   41K   83K  34% /sys/firmware/efi/efivars
/dev/sda1       511M   6.1M  505M   2% /boot/efi
tmpfs           785M  124K   785M   1% /run/user/1000
iii@III-100: ~/demo-folder$
```

26. du

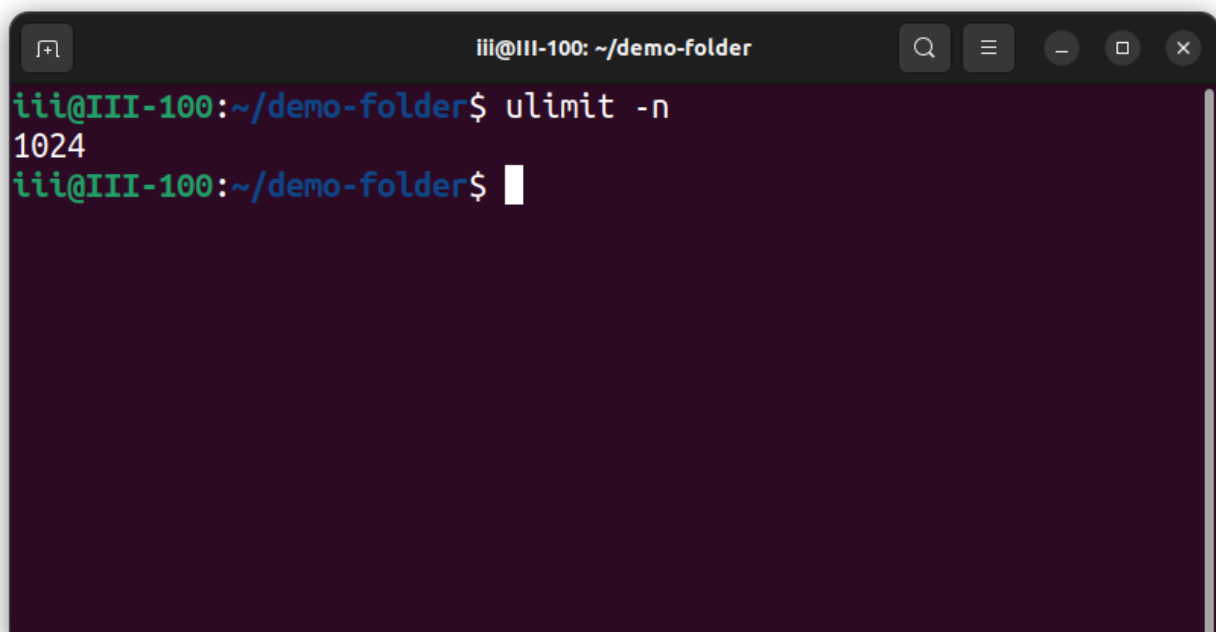
- Description: Shows disk usage of files and directories. -s gives a summary and -h uses human-readable units.
- Example: du -sh folder/

A terminal window titled 'iii@III-100: ~/demo-folder' showing the output of the 'du -sh' command. The first command 'du -sh' returns '16K .' indicating the current directory size. The second command 'du -sh demo-1/' returns '8.0K demo-1/' showing the size of the 'demo-1' subdirectory.

```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ du -sh
16K      .
iii@III-100:~/demo-folder$ du -sh demo-1/
8.0K     demo-1/
iii@III-100:~/demo-folder$
```

27. ulimit

- Description: Displays or sets shell resource limits.
- Example: ulimit -n

A terminal window titled 'iii@III-100: ~/demo-folder' showing the output of the 'ulimit -n' command. The command returns '1024', which is the current open file descriptor limit.

```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ ulimit -n
1024
iii@III-100:~/demo-folder$
```

28. cal

- Description: Displays a calendar.
- Example: cal 2025
- Example: cal 07 2025

```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ cal 8 2026
    August 2026
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
iii@III-100:~/demo-folder$
```

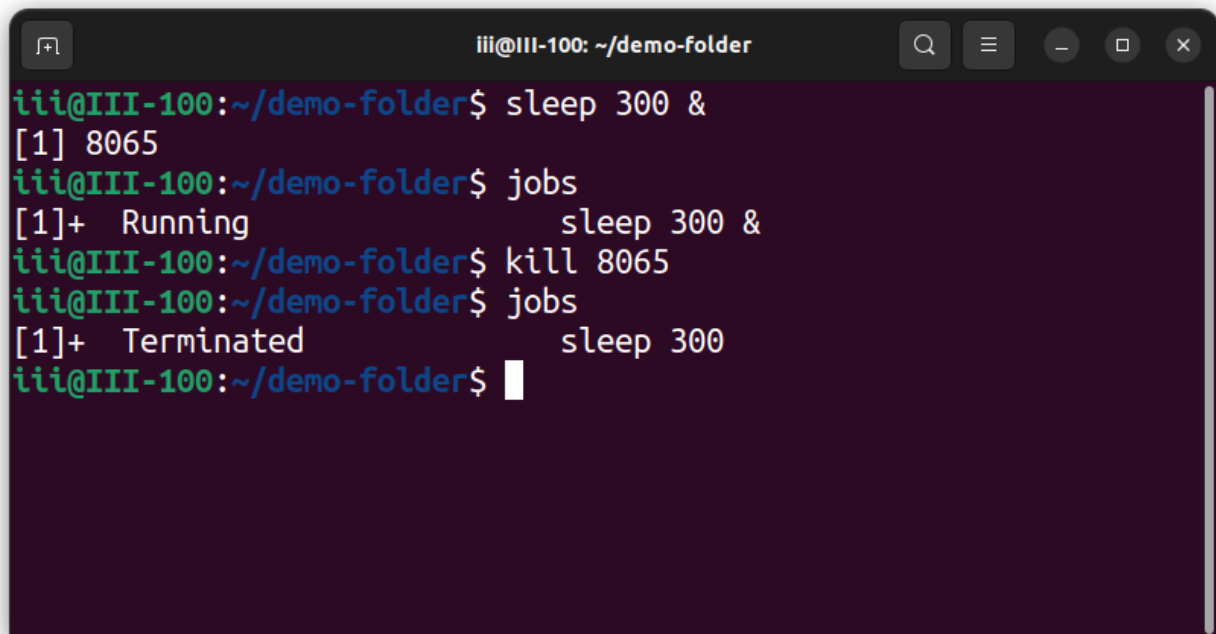
29. ps

- Description: Displays information about active processes.
- Example: ps aux

```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIM
E COMMAND
root           1  0.0  0.1 168132 13592 ?        Ss   14:23   0:0
1 /sbin/init splash
root           2  0.0  0.0      0      0 ?        S    14:23   0:0
0 [kthreadd]
root           3  0.0  0.0      0      0 ?        S    14:23   0:0
0 [pool_workqueue_release]
root           4  0.0  0.0      0      0 ?        I<   14:23   0:0
0 [kworker/R-rcu_g]
root           5  0.0  0.0      0      0 ?        I<   14:23   0:0
0 [kworker/R-rcu_p]
root           6  0.0  0.0      0      0 ?        I<   14:23   0:0
```

30. kill

- Description: Sends a signal to a process. The default signal normally requests termination.
- Example: kill 1234

A terminal window with a dark background and light-colored text. The window title is 'iii@III-100: ~/demo-folder'. The terminal shows a sequence of commands and their outputs: 1. Command: 'sleep 300 &'. Output: '[1] 8065'. 2. Command: 'jobs'. Output: '[1]+ Running sleep 300 &'. 3. Command: 'kill 8065'. 4. Command: 'jobs'. Output: '[1]+ Terminated sleep 300'. The prompt 'iii@III-100:~/demo-folder\$' is visible at the end of the last line.

```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ sleep 300 &
[1] 8065
iii@III-100:~/demo-folder$ jobs
[1]+  Running                  sleep 300 &
iii@III-100:~/demo-folder$ kill 8065
iii@III-100:~/demo-folder$ jobs
[1]+  Terminated               sleep 300
iii@III-100:~/demo-folder$
```

31. nice

- Description: Starts a process with a specified niceness value; a higher niceness generally means lower CPU scheduling priority.
- Example: nice -n 10 command

```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ sleep 300 &
[1] 8168
iii@III-100:~/demo-folder$ jobs -p %1
8168
iii@III-100:~/demo-folder$ nice -n 10 sleep 300 &
[2] 8176
iii@III-100:~/demo-folder$ jobs
[1]-  Running                sleep 300 &
[2]+  Running                nice -n 10 sleep 300 &
iii@III-100:~/demo-folder$ kill 8176
iii@III-100:~/demo-folder$
```

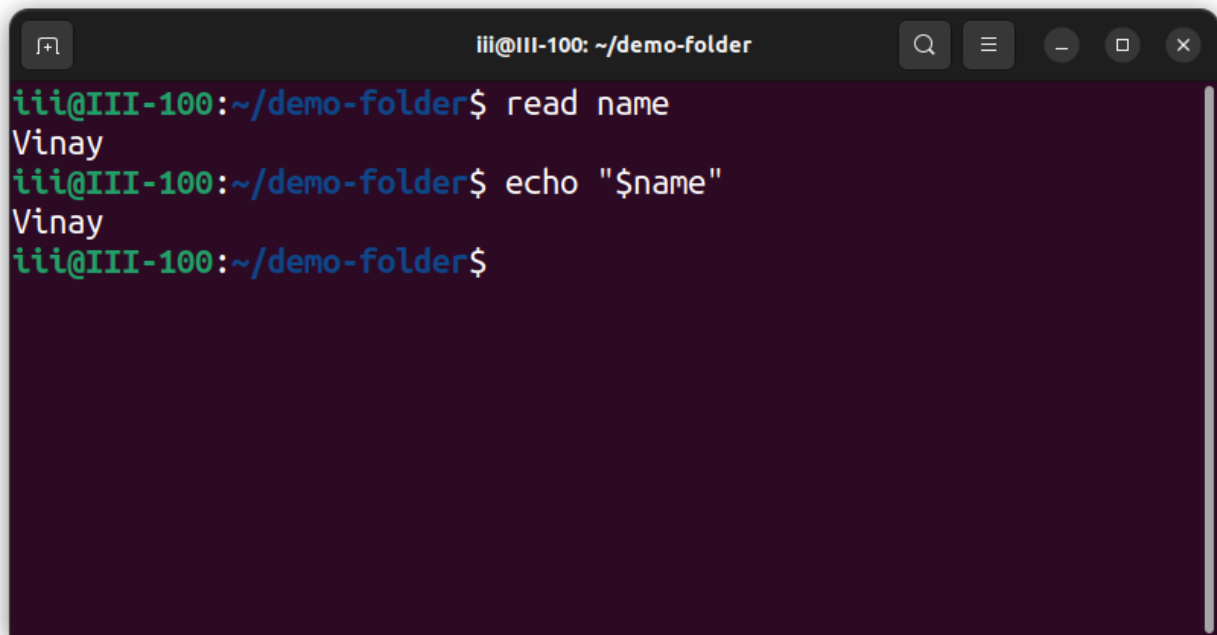
32. read

- Description: Reads input from the user and stores it in a shell variable.
- Example: read name

```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ read name
Vinay
iii@III-100:~/demo-folder$ echo "$name"
Vinay
iii@III-100:~/demo-folder$
```

33. echo

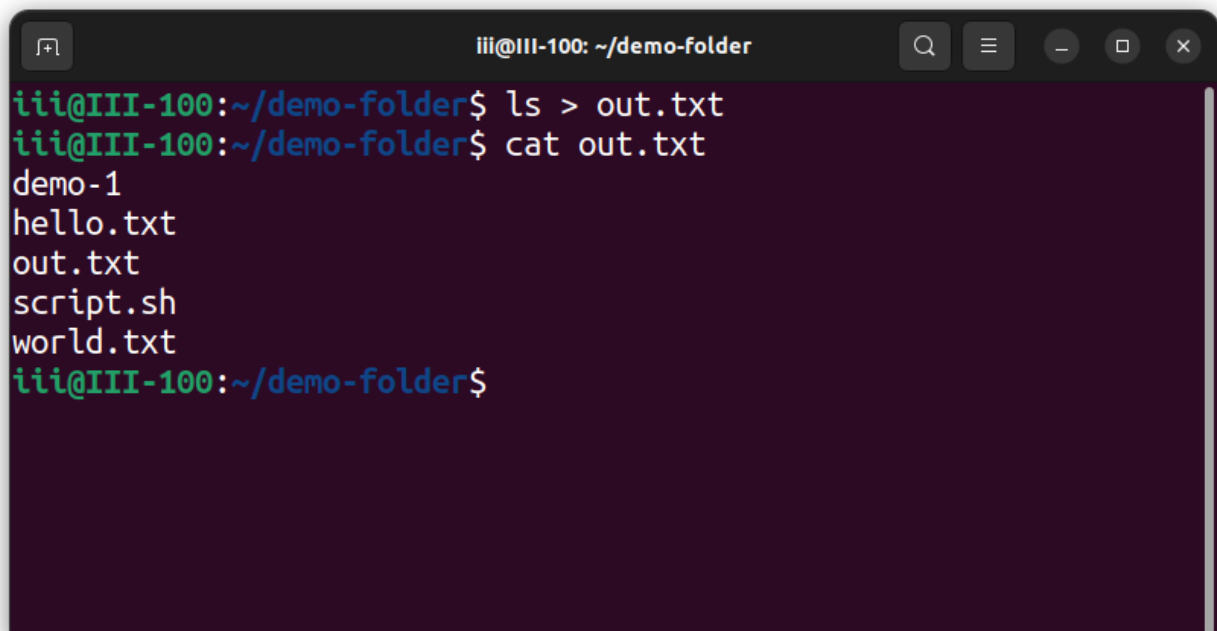
- Description: Prints text or other arguments to the terminal.
- Example: echo "Hello World"



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ read name
Vinay
iii@III-100:~/demo-folder$ echo "$name"
Vinay
iii@III-100:~/demo-folder$
```

34. I/O redirection

- Description: Redirects command input or output. > overwrites a file, >> appends to a file, and < takes input from a file.
- Example: ls > out.txt
- Example: ls >> out.txt
- Example: cat < in.txt



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ ls > out.txt
iii@III-100:~/demo-folder$ cat out.txt
demo-1
hello.txt
out.txt
script.sh
world.txt
iii@III-100:~/demo-folder$
```

35. top

- Description: Displays real-time information about processes and system resource usage.
- Example: top

```
top - 15:14:03 up 50 min, 1 user, load average: 0.18, 0.27, 0.3
Tasks: 256 total, 1 running, 255 sleeping, 0 stopped, 0 zom
%Cpu(s): 1.5 us, 2.9 sy, 0.0 ni, 95.6 id, 0.0 wa, 0.0 hi, 0
MiB Mem : 7848.5 total, 415.5 free, 2934.5 used, 4498.5
MiB Swap: 3976.0 total, 3976.0 free, 0.0 used. 4060.2
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1485	iii	20	0	5533176	290484	140012	S	12.5	3.6
7578	root	20	0	0	0	0	I	6.2	0.0
8289	iii	20	0	13212	4216	3368	R	6.2	0.1
1	root	20	0	168132	13592	8376	S	0.0	0.2
2	root	20	0	0	0	0	S	0.0	0.0
3	root	20	0	0	0	0	S	0.0	0.0
4	root	0	-20	0	0	0	I	0.0	0.0

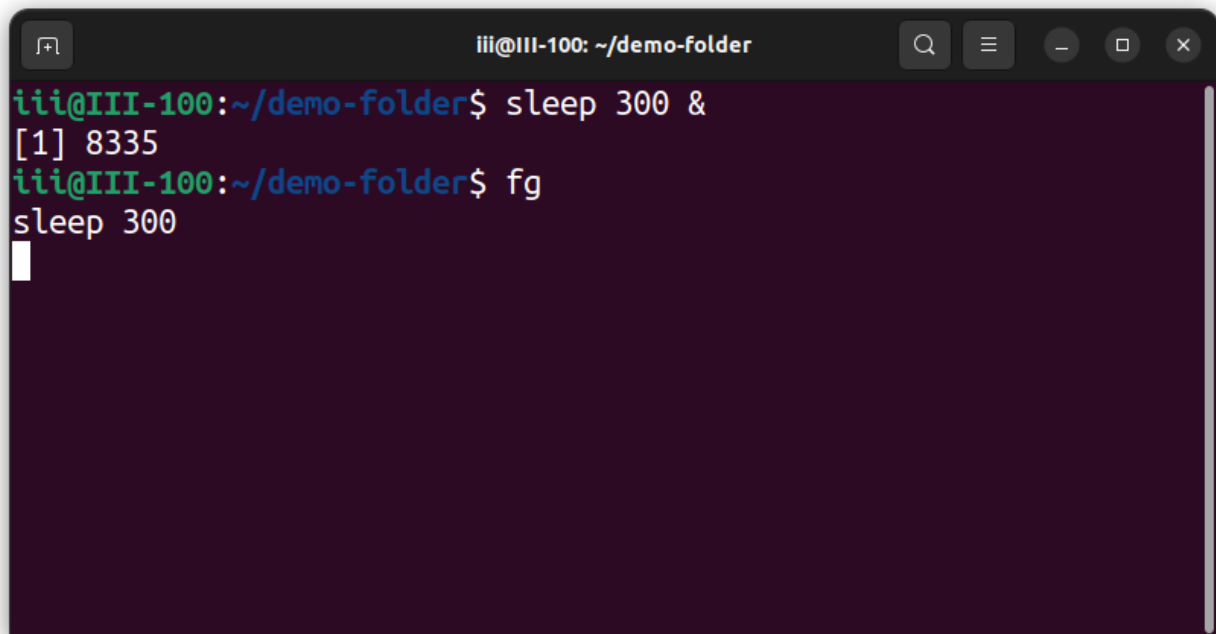
36. bg

- Description: Resumes a suspended job in the background.
- Example: bg

```
iii@III-100: ~/demo-folder$ bg
bash: bg: job 1 already in background
iii@III-100: ~/demo-folder$
```

37. fg

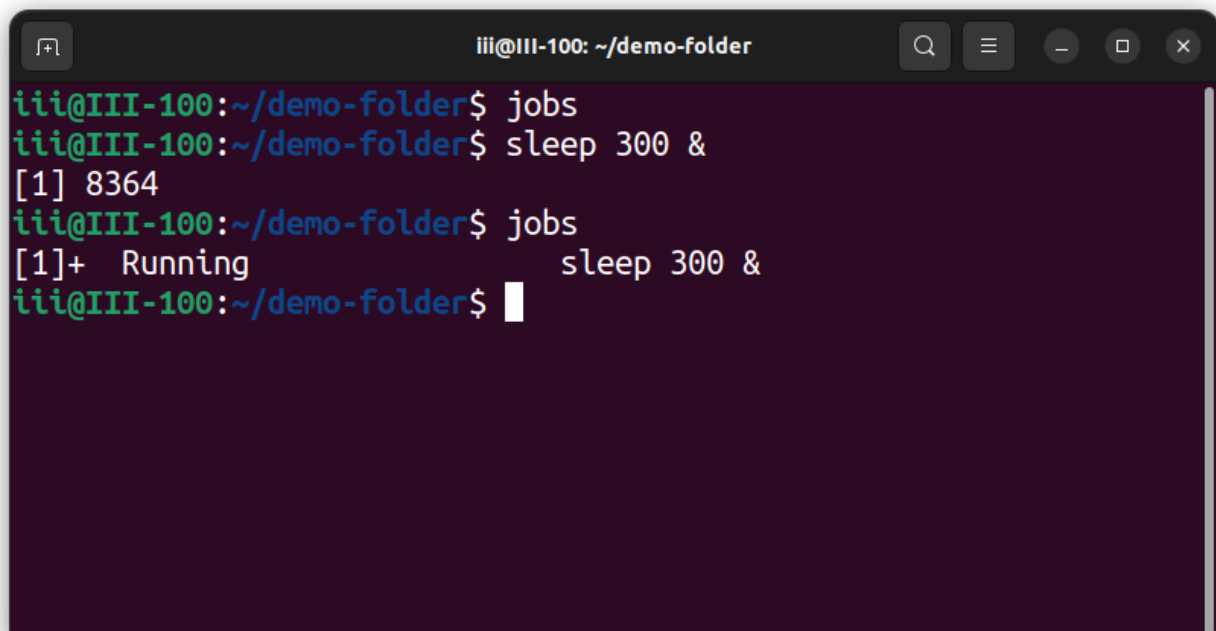
- Description: Brings a background job to the foreground.
- Example: fg



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ sleep 300 &
[1] 8335
iii@III-100:~/demo-folder$ fg
sleep 300
```

38. jobs

- Description: Lists active jobs managed by the current shell.
- Example: jobs



```
iii@III-100: ~/demo-folder
iii@III-100:~/demo-folder$ jobs
iii@III-100:~/demo-folder$ sleep 300 &
[1] 8364
iii@III-100:~/demo-folder$ jobs
[1]+  Running                  sleep 300 &
iii@III-100:~/demo-folder$
```